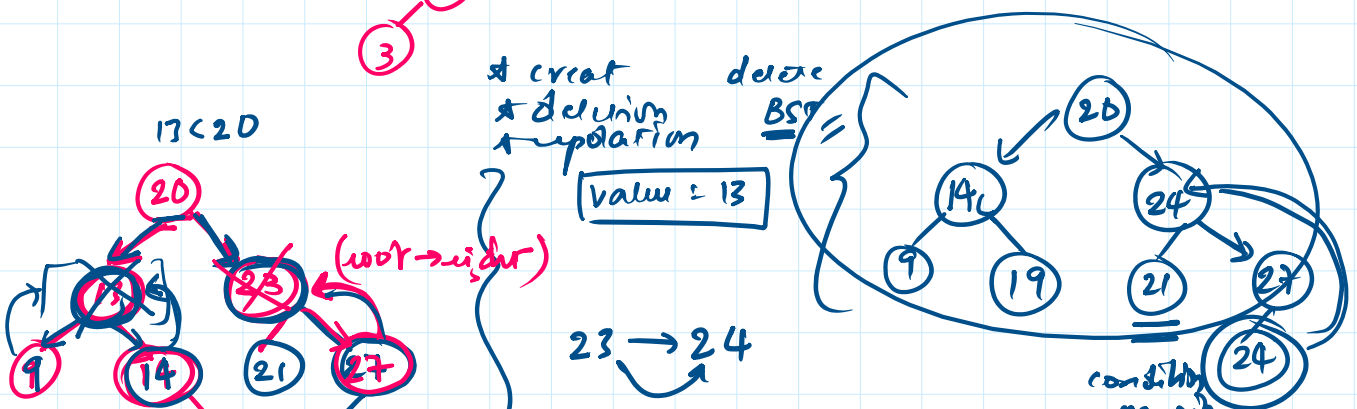
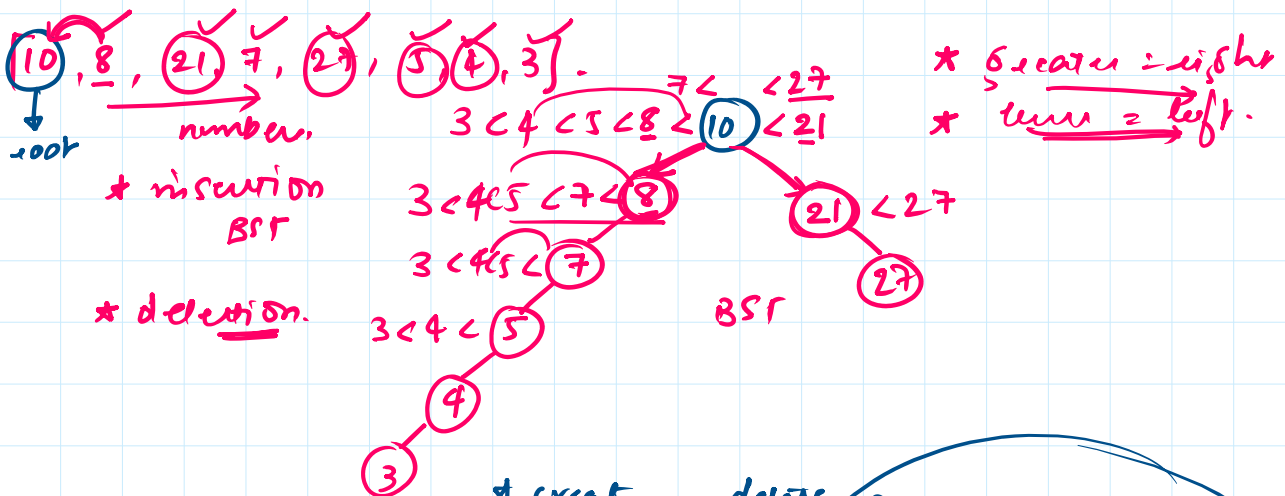
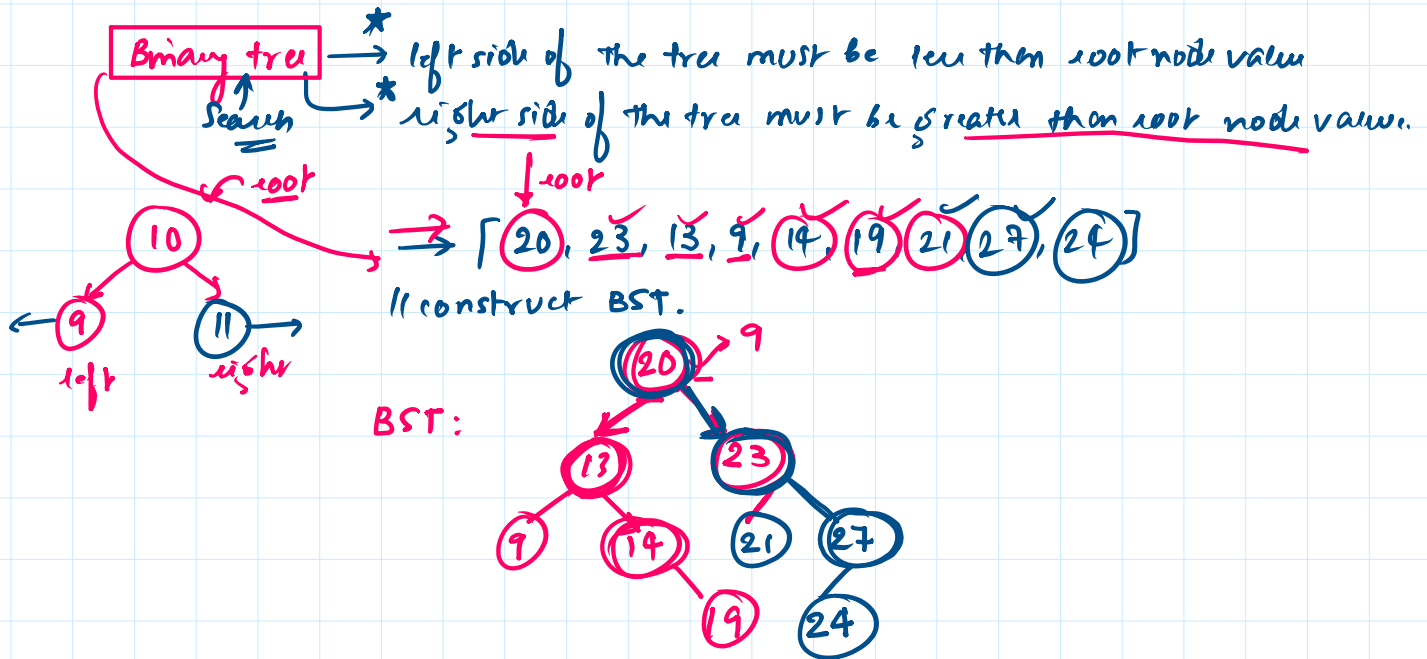
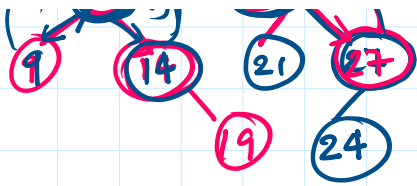


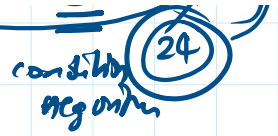
* Binary tree: Algorithm $\rightarrow$ Array $\rightarrow$ find an element in $O(\log n)$
 * **Binary search** trees
 linear search $\rightarrow O(n)$

array $\rightarrow$ Binary search $\rightarrow$ Arr 17 $\rightarrow$ Sorted [Asc $\rightarrow$ desc]





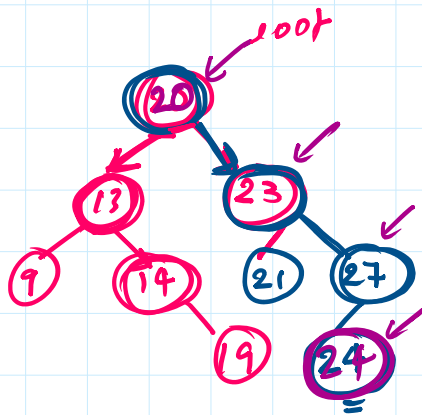
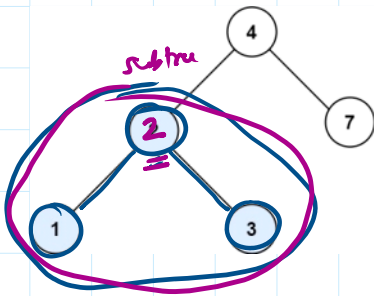
23 → 24



+ search in BST → LC
~~+~~ insert in BST → LC

if (root → value == value) {
 return root;

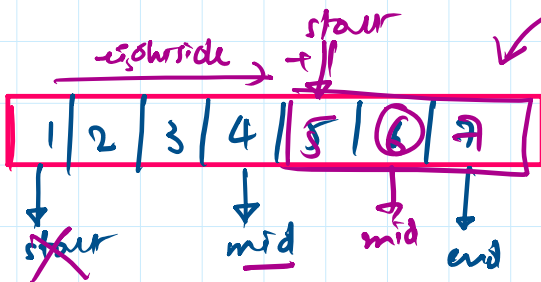
value = 2
 → true
 → true



value = 24
 search
 if (root → value == value) {
 return root;

20 < 24
 if (root → value < value) {
 // move towards right side.

Else // move towards left side.

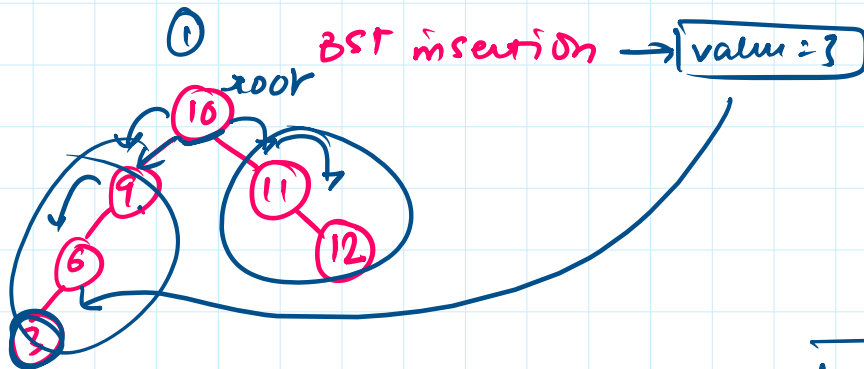


if (arr[mid] == key) {
 return true;
 }
 if (arr[mid] < 6) {
 4 <

$$\text{if } (\text{arr[mid]} < 6) \{$$

$$4 <$$

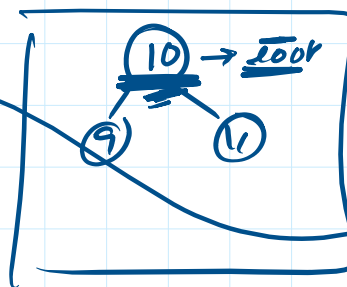
$$\text{start} = \text{mid} + 1$$



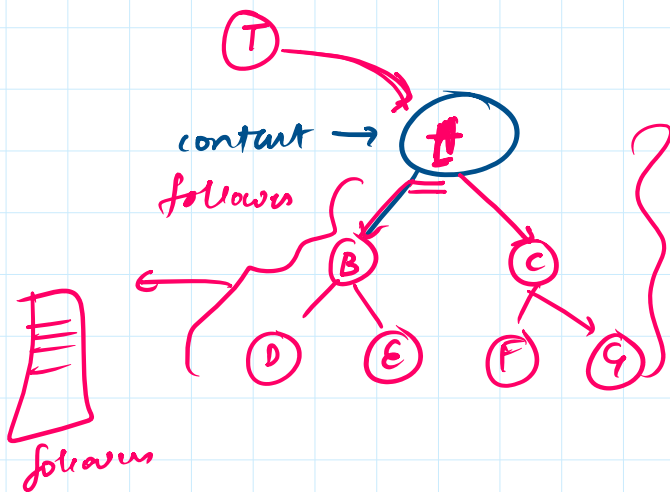
$\text{root} \rightarrow \text{left} < \text{value}$
 $\text{root} \rightarrow \text{right} > \text{value}$

②

new TreeNode(value)



$\text{if } (\text{root} == \text{NULL}) \{$
 return
 new TreeNode(value);



BST → research Application BST
 ↳ use cases...